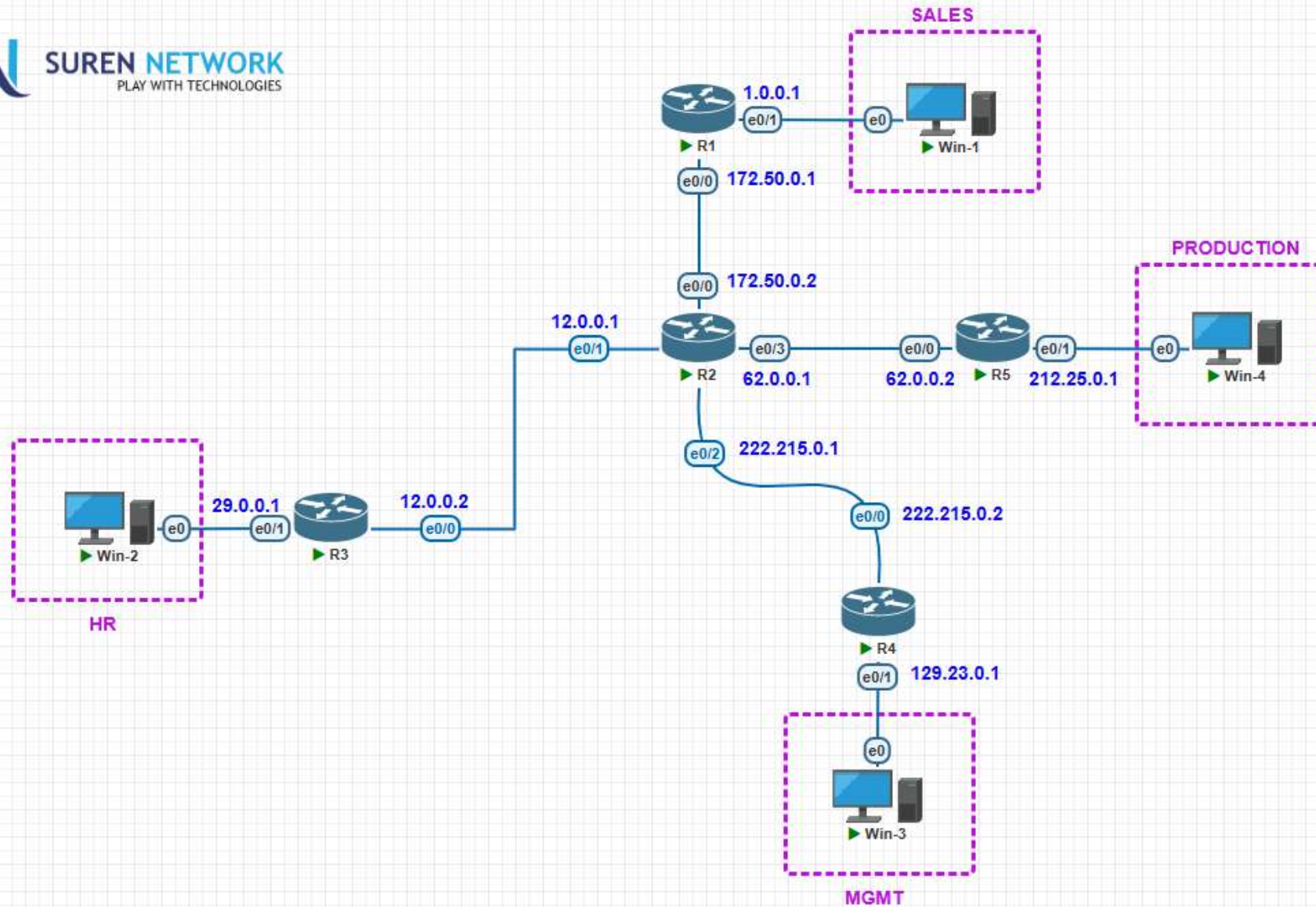


TOPOLOGY FOR DHCP

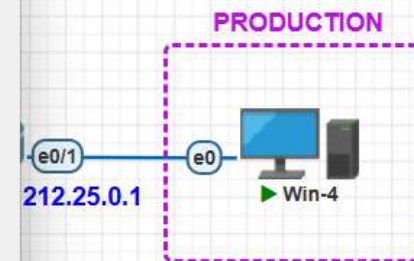
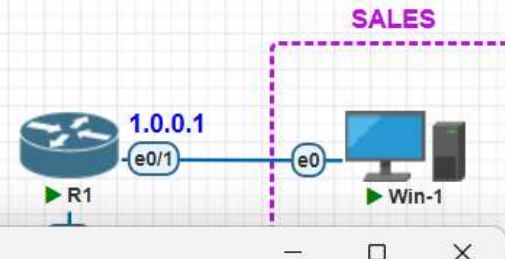


Router Config of DHCP & Department Pool & Routing

```
R2(config)#int e0/0
R2(config-if)#ip add 172.50.0.2 255.255.0.0
R2(config-if)#no sh
R2(config-if)#int e0/1
R2(config-if)#ip add 12.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#int e0/2
R2(config-if)#ip add 222.215.0.1 255.255.255.0
R2(config-if)#no sh
R2(config-if)#int e0/3
R2(config-if)#ip add 62.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#ip dhcp pool HR
R2(dhcp-config)#network 29.0.0.0 255.0.0.0
R2(dhcp-config)#default-router 29.0.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp excluded-address 29.0.0.1 29.0.0.10
R2(config)#ip dhcp pool SALES
R2(dhcp-config)#network 1.0.0.0 255.0.0.0
R2(dhcp-config)#default-router 1.0.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp excluded 1.0.0.1 1.0.0.10
R2(config)#
*Jul 29 08:36:33.440: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config)#ip dhcp pool PRODUCTION
R2(dhcp-config)#network 212.25.0.0 255.255.255.0
R2(dhcp-config)#default-router 212.25.0.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp excluded-address 212.25.0.1 212.25.0.10
R2(config)#ip dhcp pool MGMT
R2(dhcp-config)#network 129.23.0.0 255.255.0.0
R2(dhcp-config)#default-router 129.23.0.1
R2(dhcp-config)#dns-sever 8.8.8.8
      ^
% Invalid input detected at '^' marker.

R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#exit
R2(config)#ip dhcp excluded-address 129.23.0.1 129.23.0.10
R2(config)#exit
R2#
*Jul 29 08:40:33.947: %SYS-5-CONFIG_I: Configured from console by console
R2#conf t
```

DHCP Table - After Assigning the IP for the Device



```
R2
% Invalid input detected at '^' marker.

R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip dhcp binding
% Incomplete command.

R2(config)#exit
R2#
*Jul 29 09:03:58.313: %SYS-5-CONFIG_I: Configured from console by console
R2#
R2#
R2#
R2#
R2#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type
Hardware address/
User name
1.0.0.11        0150.b7bb.06ac.00  Jul 30 2026 08:58 AM Automatic
29.0.0.11       0150.cd1b.06ad.00  Jul 30 2026 08:58 AM Automatic
129.23.0.11     0150.ce12.06ae.00  Jul 30 2026 08:59 AM Automatic
212.25.0.11     0150.6531.06af.00  Jul 30 2026 08:58 AM Automatic
R2#
```



Remaining Router Routing Configuration

```
R3
2, changed state to down
*Jul 29 08:44:05.145: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
R3(config)#int e0/0
R3(config-if)#ip add 12.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#int e0/1
R3(config-if)#ip add 29.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#
R3(config-if)#exit
R3(config)#ip route 1.0.0.0 255.0.0.0 12.0.0.1
R3(config)#ip route 62.0.0.0 255.0.0.0 12.0.0.1
R3(config)#ip route 172.50.0.0 255.255.0.0 12.0.0.1
R3(config)#ip route 212.25.0.0 255.255.255.0 12.0.0.1
R3(config)#ip route 222.215.0.0 255.255.255.0 12.0.0.1
R3(config)#ip route 129.23.0.0 255.255.0.0 12.0.0.1
R3(config)#int e0/1
R3(config-if)#
*Jul 29 08:48:53.791: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R3(config-if)#ip helper-address 12.0.0.1
R3(config-if)#exit
R3(config)#
```

```
R5#
R5#
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#int e0/0
R5(config-if)#ip add 62.0.0.2 255.0.0.0
R5(config-if)#no sh
R5(config-if)#int e0/1
R5(config-if)#ip add 212.25.0.1 255.255.255.0
R5(config-if)#ip helper-address 62.0.0.1
R5(config-if)#exit
R5(config)#ip rout
*Jul 29 08:55:39.642: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R5(config)#ip route
% Incomplete command.

R5(config)#ip route 129.23.0.0 255.255.0.0 62.0.0.1
R5(config)#ip route 222.215.0.0 255.255.255.0 62.0.0.1
R5(config)#ip route 12.0.0.0 255.0.0.0 62.0.0.1
R5(config)#ip route 29.0.0.0 255.0.0.0 62.0.0.1
R5(config)#ip route 1.0.0.0 255.0.0.0 62.0.0.1
R5(config)#ip route 172.50.0.0 255.255.0.0 62.0.0.1
R5(config)#
```

```
R1
R1(config)#
R1(config)#
R1(config)#
R1(config)#int e0/0
R1(config-if)#ip add 172.50.0.1 255.255.0.0
R1(config-if)#no sh
R1(config-if)#int e0/1
R1(config-if)#ip add 1.0.0.1 255.0.0.0
R1(config-if)#ip helper-address 172.50.0.2
R1(config-if)#exit
R1(config)#ip route 212.25.0.0 255.255.255.0 172.50.0.2
R1(config)#ip route 62.0.0.0 255.0.0.0 172.50.0.2
R1(config)#ip route 222.215.0.0 255.255.255.0 172.50.0.2
R1(config)#ip route 129.23.0.0 255.255.0.0 172.50.0.2
R1(config)#ip route 12.0.0.0 255.0.0.0 172.50.0.2
^
% Invalid input detected at '^' marker.

R1(config)#ip route 12.0.0.0 255.0.0.0 172.50.0.2
R1(config)#ip route 29.0.0.0 255.0.0.0 172.50.0.2
R1(config)#
*Jul 29 08:55:31.684: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R1(config)#
```

```
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int e0/0
R4(config-if)#ip add 222.215.0.2 255.255.255.0
R4(config-if)#no sh
R4(config-if)#int e0/1
R4(config-if)#ip add 129.23.0.1 255.255.0.0
R4(config-if)#ip helper-address 222.215.0.1
R4(config-if)#exit
R4(config)#ip route 29.0.0.0 255.0.0.0 222.215.0.1
R4(config)#ip route 12.0.0.0 255.0.0.0 222.215.0.1
R4(config)#ip route 1.0.0.0 255.0.0.0 222.215.0.1
R4(config)#ip route 62.0.0.0 255.0.0.0 222.215.0.1
R4(config)#ip route 172.50.0.0 255.255.0.0 222.215.0.1
R4(config)#ip route 212.25.0.0 255.255.255.0 22.215.0.1
^
% Invalid input detected at '^' marker.

R4(config)#ip route 212.25.0.0 255.255.255.0 22.215.0.1
R4(config)#
*Jul 29 09:02:19.024: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
R4(config)#
```

Successfully IP Address was Assigned by DHCP

